

1. 设独立随机变量 $X_1, \dots, X_n$ 服从如下分布

$$P(X_i \leq x | \alpha, \beta) = \begin{cases} 0 & x < 0 \\ (x/\beta)^\alpha & 0 \leq x < \beta \\ 1 & x > \beta \end{cases}$$

(a) 求 α, β 的矩估计量;

(b) 求 α, β 的最大似然估计;

(c) 在篱雀的巢中找到的杜鹃蛋的长度 (单位: mm) 可以用这个分布建模。根据如下表格中的数据, 计算出 (a)

(b) 中的估计量的观测值。

22.0	23.9	20.9	23.8	25.0	24.0	21.7	23.8	22.8	23.1	23.1	23.5	23.0	23.0
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$$(a) \quad f(x) = \frac{d}{dx} F(x) = \frac{\alpha}{\beta} \left(\frac{x}{\beta}\right)^{\alpha-1} I_{[0, \beta]}(x)$$

$$E(x) = \int_0^\beta f(x) x dx = \int_0^\beta \frac{\alpha}{\beta^\alpha} x^\alpha dx = \frac{\alpha}{(\alpha+1)\beta^\alpha} \beta^{\alpha+1} = \frac{\alpha\beta}{\alpha+1} \quad ①$$

$$E(x^2) = \int_0^\beta f(x) x^2 dx = \int_0^\beta \frac{\alpha}{\beta^\alpha} x^{\alpha+1} dx = \frac{\alpha}{(\alpha+2)\beta^\alpha} \beta^{\alpha+2}$$

$$\begin{aligned} \text{Var}(x) &= E(x^2) - E^2(x) = \left(\frac{\alpha}{\alpha+2} - \left(\frac{\alpha}{\alpha+1}\right)^2\right) \beta^2 \\ &= \frac{\alpha\beta^2}{(\alpha+1)(\alpha+2)} \quad ② \end{aligned}$$

联立①② 解得

$$\alpha = \sqrt{\frac{E^2(x)}{\text{Var}(x)} + 1} - 1 \quad \beta = \frac{\text{Var}(x) \left(\sqrt{\frac{E^2(x)}{\text{Var}(x)} + 1} + 1 \right)}{E(x)} + E(x)$$

$$\hat{\alpha} = \sqrt{\frac{\bar{x}^2}{S(x)} + 1} - 1 \quad \hat{\beta} = \frac{S(x) \left(\sqrt{\frac{\bar{x}^2}{S(x)} + 1} + 1 \right)}{\bar{x}} + \bar{x}$$

$$(b) \quad f_{\alpha, \beta}(\vec{x}) = \left(\frac{\alpha}{\beta^\alpha}\right)^n (x_1 \cdots x_n)^{(\alpha-1)}$$

$$l(\alpha, \beta) = n \log \alpha - n \alpha \log \beta + \sum_{i=1}^n n(\alpha-1) \log x_i$$

$$\begin{cases} \frac{\partial \ell}{\partial \alpha} = 0 \\ \frac{\partial \ell}{\partial \beta} = 0 \end{cases} \Rightarrow \begin{cases} \frac{n}{\alpha} - n \log \beta + \sum_{i=1}^n \log x_i = 0 \quad ① \\ \frac{n\alpha}{\beta} = 0 \end{cases}$$

$\alpha=0$ 不能取到, 由 $\alpha>0$ 知 $\ell(\alpha, \beta)$ 随 β 增大而减小, 因此 $\hat{\beta}_{MLE} = \max(X_1, \dots, X_n) = X_{(n)}$

代入①中得 $\alpha = \frac{n}{-\sum_{i=1}^n \log x_i + n \log \hat{\beta}} = \frac{n}{-\sum_{i=1}^n \log x_i + n \log x_{(n)}}$

(C) $\begin{cases} \hat{\alpha}_{MM} = 21.05 \\ \hat{\beta}_{MM} = 24.12 \end{cases} \quad \begin{cases} \hat{\alpha}_{MLE} = 12.59 \\ \hat{\beta}_{MLE} = 25.0 \end{cases}$

2. 设独立随机变量 $X_1, \dots, X_n$ 独立同分布且具有概率密度函数

$$f(x|\theta) = \theta x^{\theta-1}, 0 \leq x \leq 1, 0 < \theta < \infty$$

(a) 求 θ 的最大似然估计, 并证明当 $n \rightarrow \infty$ 时, 它的方差 $\rightarrow 0$ 。

(b) 求 θ 的矩估计量。

$$(a) f(\vec{x}|\theta) = \theta^n (x_1 \cdots x_n)^{\theta-1} \prod_{i=1}^n I_{(0,1)}(x_i)$$

$$\ell(\theta) = n \log \theta + \sum_{i=1}^n (\theta-1) \log x_i$$

$$\frac{\partial \ell}{\partial \theta} = 0 \Rightarrow \frac{n}{\theta} + \sum_{i=1}^n \log x_i = 0 \Rightarrow \theta = -\frac{n}{\sum_{i=1}^n \log x_i}$$

$$\hat{\theta}_{MLE} = \frac{-n}{\sum_{i=1}^n \log x_i}$$

$$\text{令 } Y = -\log X \quad f(y) = f(x) \frac{dx}{dy} = \theta x^\theta = \theta e^{-y\theta} I_{(0,\infty)}(y)$$

$$Y \sim \text{Exp}(\theta) \quad E(Y) = \frac{1}{\theta} \quad \text{Var}(Y) = \frac{1}{\theta^2}$$

$$\hat{\theta} = \frac{1}{\bar{Y}} \quad \bar{Y} \sim \Gamma(1, n\theta)$$

$$\text{由Gamma分布性质} \quad \text{Var}\left(\frac{1}{\bar{Y}}\right) = \frac{n^2 \theta^2}{(n-1)^2 (n-2)}$$

$$\lim_{n \rightarrow \infty} \text{Var}\left(\frac{1}{\bar{Y}}\right) = 0$$